

Electricity & Magnetism

Part II practice session

- §1 — Special relativity
- §2 — Field transformations
- §3 — Potentials, gauges
- §4 — E&M in index notation
- §5 — Radiation
- §6 — Waves in matter

§1 Relativity



$$A = (A^0, A^1, A^2, A^3)$$

4-vector if in S' :

$$A' = (A'^0, A'^1, A'^2, A'^3)$$

$$A'^0 = \gamma(A^0 - v A^1)$$

$$A'^1 = \gamma(A^1 - v A^0)$$

$$A'^2 = A^2, A'^3 = A^3$$

$$A' = \Lambda A \quad (4 \text{ equations})$$

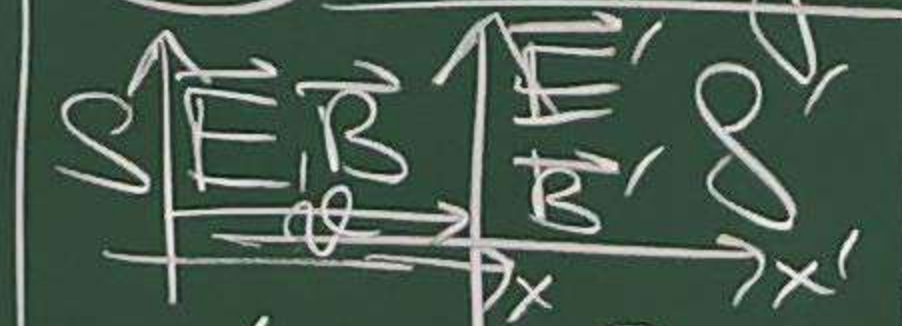
$$A'^M = (\Lambda^M_{\nu}) \cdot A^\nu$$

$$= \sum_{\nu=0,1,2,3} \Lambda^M_{\nu} A^\nu$$

$$= \Lambda^M_{\nu} A^\nu$$

$$\Lambda = \begin{pmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

§2 Field trans



$$E'_x = E_x$$

$$B'_x = B_x$$

$$E'_y = \gamma(E_y - v B_z)$$

$$B'_y = \gamma(B_y + \frac{v}{c^2} E_z)$$

$$E'_z = \gamma(E_z + v B_y)$$

$$B'_z = \gamma(B_z - \frac{v}{c^2} E_y)$$

$$x' = x$$

$$y' = y$$

$$z' = z$$

$$t' = \gamma(t - \frac{v}{c^2} x)$$

$$t = \gamma(t' + \frac{v}{c^2} x')$$

$$x = \gamma(x' + vt')$$

$$t = \gamma(t' + \frac{v}{c^2} x')$$

① rest frame

$$\vec{E} \text{ only}$$

$$\vec{B} = 0$$

in another frame

$$E'_x = E_x, E'_y = \gamma E_y$$

$$E'_z = \gamma E_z$$

$$A = A^M = \begin{pmatrix} V/c \\ \vec{A} \end{pmatrix}$$

$$\vec{E} = 0$$

$$\vec{E}' = \vec{v} \times \vec{B}$$

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\nabla V - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

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$$\vec{B} = \nabla \times \vec{A}$$

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$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\nabla V - \frac{\partial \vec{A}}{\partial t}$$

Maxwell's for fields

$$\vec{E} = \vec{E}^M = \begin{pmatrix} V/c \\ \vec{E} \end{pmatrix}$$

$$\vec{B} = \vec{B}^M = \begin{pmatrix} 0 \\ \vec{B} \end{pmatrix}$$

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② 4 equations for potentials

$$\square^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0} \rho$$

$$\square^2 \vec{A} - \nabla L = -\mu_0 \vec{J}$$

$$\square^2 = \nabla^2 - \frac{1}{c^2} \frac{\partial^2}{\partial t^2}$$

$$L = \nabla \cdot \vec{A} + \mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

$$\square \cdot \vec{A} = \square_\mu A^\mu$$

$$\square^2 = \square \cdot \square$$

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$$\square^2 = \square \cdot \square$$

① Coulomb

$$\vec{\nabla} \cdot \vec{A} = 0$$

NOT Lorentz invariant

useful for (quasi)-static

$$\vec{E} = -\nabla V$$

$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\nabla V$$

$$\vec{B} = \nabla \times \vec{A}$$

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$$\vec{B} = \nabla \times \vec{A}$$

$$\vec{E} = -\nabla V$$

② Lorentz gauge

$$\square \cdot \vec{A} = 0$$

Lorentz-invariant

all frames

$$\square \cdot \vec{A} = 0$$

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Freedom (gauge):

$$\begin{aligned} \vec{A} &\rightarrow \vec{A} + \vec{\nabla} \chi \\ \chi(x) &\text{ scalar} \\ \vec{V} &\rightarrow \vec{V} - \frac{\partial \chi}{\partial t} \end{aligned}$$

→ eqns for $\vec{E}, \vec{B}$ stay the same {verify!}

(?) Find 1 for $\odot$ and $\otimes$!

(34) Index nott $\eta^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$

$$\begin{aligned} \partial_\mu A^\mu &= \left(\frac{1}{c} \frac{\partial}{\partial t} \right) \cdot \left(\frac{V}{c} \right) \\ &= \partial_\mu A^\mu = \eta_{\mu\nu} \partial^\mu A^\nu = \left(\frac{1}{c} \frac{\partial}{\partial t} \right) \cdot \left(\frac{V}{c} \right) \end{aligned}$$

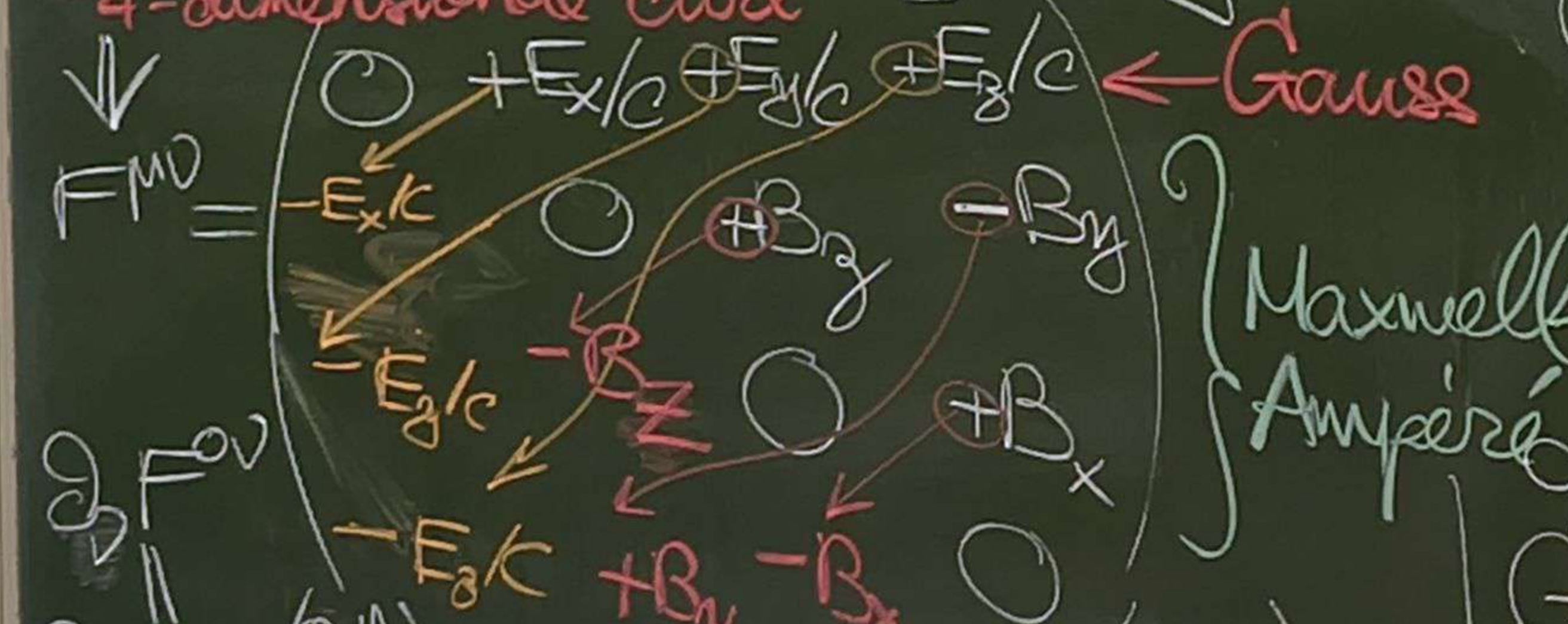
$$\eta: \mathbb{R}^1 \times \mathbb{R}^3 \rightarrow \mathbb{R}$$

$(\partial_\mu, \omega) \mapsto \eta_{\mu\nu} \partial^\mu \omega^\nu$

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$$

"4-dimensional curl"

when we take divergence $\rightarrow \odot$



$$\partial_\mu F^{\mu\nu} = \frac{1}{c} \frac{\partial}{\partial t} \left(\frac{V}{c} \right) + \vec{\nabla} \cdot \left(\frac{\vec{E}}{c} \right)$$

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} = \frac{\rho}{\epsilon_0} \cdot \frac{1}{c^2} = \frac{\rho}{\epsilon_0 c^2} = \mu_0 J^0$$

Maxwell's eq.

$$\partial_\nu F^{\mu\nu} = \mu_0 J^\mu$$

dual tensor

$$G^{\mu\nu} \quad \left| \quad \frac{E_i}{c} \leftrightarrow B_i \right|$$

$$\partial_\nu G^{\mu\nu} = 0$$

Dirac $\vec{\nabla} \cdot \vec{B} = 0$ Faraday

$$\vec{E} = \vec{\nabla} \phi = \mu_0 \frac{\partial \vec{B}}{\partial t} + \mu_0 \frac{\partial \vec{B}}{\partial t} \quad \frac{F}{A} = \rho \cdot \frac{1}{c} = \frac{I}{c}$$

$$\vec{B} = \vec{B}_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)} \rightarrow \vec{k}^2 = [\mu_0 \epsilon_0 \omega^2] + \frac{1}{c^2}$$

$$\vec{Q} = \frac{1}{2} + iK \quad i[\mu_0 \epsilon_0]$$

mgk.ac/teach/mva.pdf

Remark 20.7

$$\begin{cases} F = dA \Rightarrow dF = 0 \\ d^*F = \mu_0 J \end{cases}$$